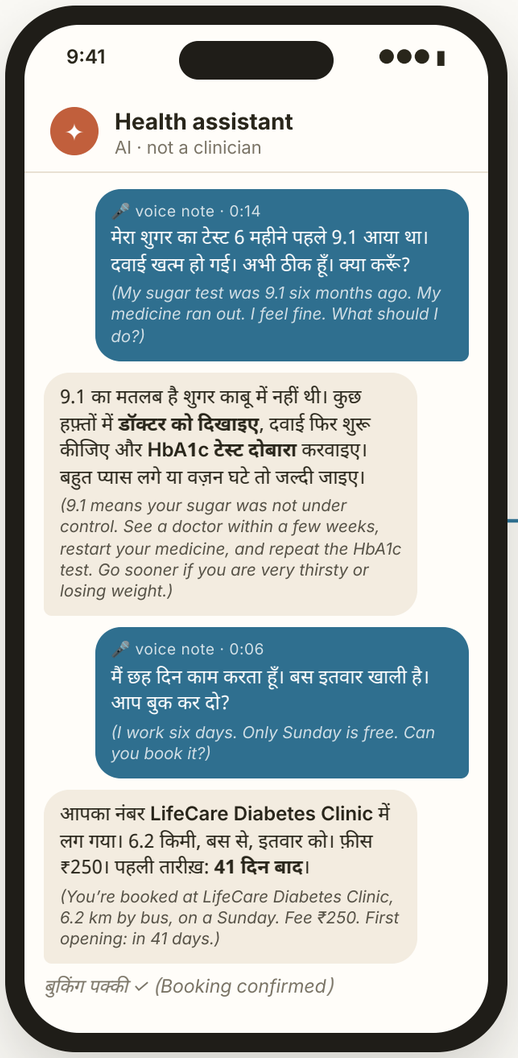
From a good answer to a reachable appointment

A 40-year-old factory worker in a Haryana village asks an AI health app about overdue diabetes care. The answer is right. Whether he gets care depends on where the app routes him.

1 · THE ANSWER



2 · THE FRONT DOOR TO CARE

The app now chooses where he goes.

AI as router

It searches a directory, ranks options, shows a few, and books one.

Home	village, 15 km limit
Coverage	ESI card, ₹300 to spare
Work	six-day factory shifts
Transport	bus and shared auto
Deadline	seen within 30 days

DESTINATION

Doctor's review with a repeat HbA1c test, to restart medicines and plan follow-up

3 · WHERE THE ROUTE CAN BREAK

c1 Unreachable from where he lives — × Too far
A sponsored diabetes centre in the city is 34 km away. He can travel 15.

c2 Outside his coverage — × Not covered
A private hospital 9 km away does not take his ESI card. Self-pay: ₹1,800, about three days' wages.

c3 No doctor or HbA1c test nearby — × No doctor or test
The village health centre checks finger-prick sugar but has no doctor on most days and no HbA1c test.

c4 Can't get time off work — × No Sunday/evening
The ESI dispensary is covered and 11 km away, but open 8 am to 2 pm, Monday to Saturday. A missed shift costs ₹550.

c5 No bus to the clinic — × No transit
The community health centre is nearly free and open Sundays, but no bus goes there. Shared autos stop 4 km short.

c6 Booked, but too late — × Past deadline
The app's partner clinic, ranked first. Close, affordable, on a bus line, but the first Sunday slot is 41 days out. The app books it anyway.

c8 A route that works, shown but passed over — ✓ Would work
A charity diabetes clinic 7.5 km away by bus, evening hours, ₹100, slot in 5 days. It was third on screen, below the partner clinic the app booked.

4 · THE AUDIT, WORKED OUT FOR HIM

FEASIBLE SET: THE CLINICS THAT WOULD ACTUALLY WORK

$$F_i = \{\ c \in S_i : d(c, h_i) \le m_i,\ \ w_c \le \tau,\ \ \pi_i(c) \le b_i,\ \ a_i(c) = 1\ \}$$

S_i	clinics offering what he needs: a doctor's review with an HbA1c test
$d(c, h_i) \le m_i$	distance from his home within his travel limit, 15 km
$w_c \le \tau$	next appointment within the clinical deadline, 30 days for overdue follow-up
$\pi_i(c) \le b_i$	his out-of-pocket cost within budget: ₹0 at ESI clinics, the full fee anywhere else
$a_i(c) = 1$	meets his stated needs: reachable by bus or auto, and open Sundays or evenings

For him: $m = 15$ km, $\tau = 30$ days, $b =$ ₹300, $a =$ bus or auto ∧ Sunday or evening hours.

WHAT THE APP DID, SCORED AGAINST F

$$R = |C \cap F| \,/\, |F| \qquad D = |L \cap F| \,/\, |F| \qquad E = 1[\mathit{booked} \in F]$$

C = clinics the search retrieved, L = clinics shown to him, E = the booking works

CLINIC	DOCTOR + HBA1C	KM ≤ 15	DAYS TO APPT ≤ 30	HIS COST ≤ ₹300	BUS OR AUTO	HOURS SUN/EVE	IN F?	SEARCH RANK	SHOWN (L)
c1 Apex Diabetes Centre (sponsored)	yes	34.0	2	₹1,400	yes	Sun	no	2	✓
c2 Sunrise Multispeciality (sponsored)	yes	9.0	1	₹1,800	yes	Eve	no	4	
c3 Village Health Centre	no	2.1	·	·	·	·	no	not listed	
c4 ESI Dispensary	yes	11.0	4	₹0	yes	weekdays	no	5	
c5 Community Health Centre	yes	13.5	6	₹10	no	Sun	no	9 · p.2	
c6 LifeCare Diabetes (partner) BOOKED	yes	6.2	41	₹250	yes	Sun	no	1	✓
c7 ESI Sunday Clinic	yes	12.4	8	₹0	yes	Sun	yes	7 · p.2	
c8 Seva Trust Diabetes Clinic	yes	7.5	5	₹100	yes	Eve	yes	3	✓

$F = \{c7, c8\},\ C = \{c6, c1, c8, c2, c4\}$
$L = \{c6, c1, c8\}$
Page 1 of the search; top 3 shown.

$R = |\{c8\}| / 2 = 0.5$
$D = |\{c8\}| / 2 = 0.5$
Found and showed one of two.

$E = 1[c6 \in F] = 0$
Answer correct, route failed: he waits 41 days.

WHAT THE AUDIT ASKS

Grading the answer alone sees none of step 3.

Found it? Did the search retrieve the routes that work for him? RETRIEVAL, R

Showed it? Did he see a usable option, and was it ranked where he would pick it? DISPLAY, D

Booked it? Does the booking meet his deadline, budget, travel, and work limits? EXECUTION